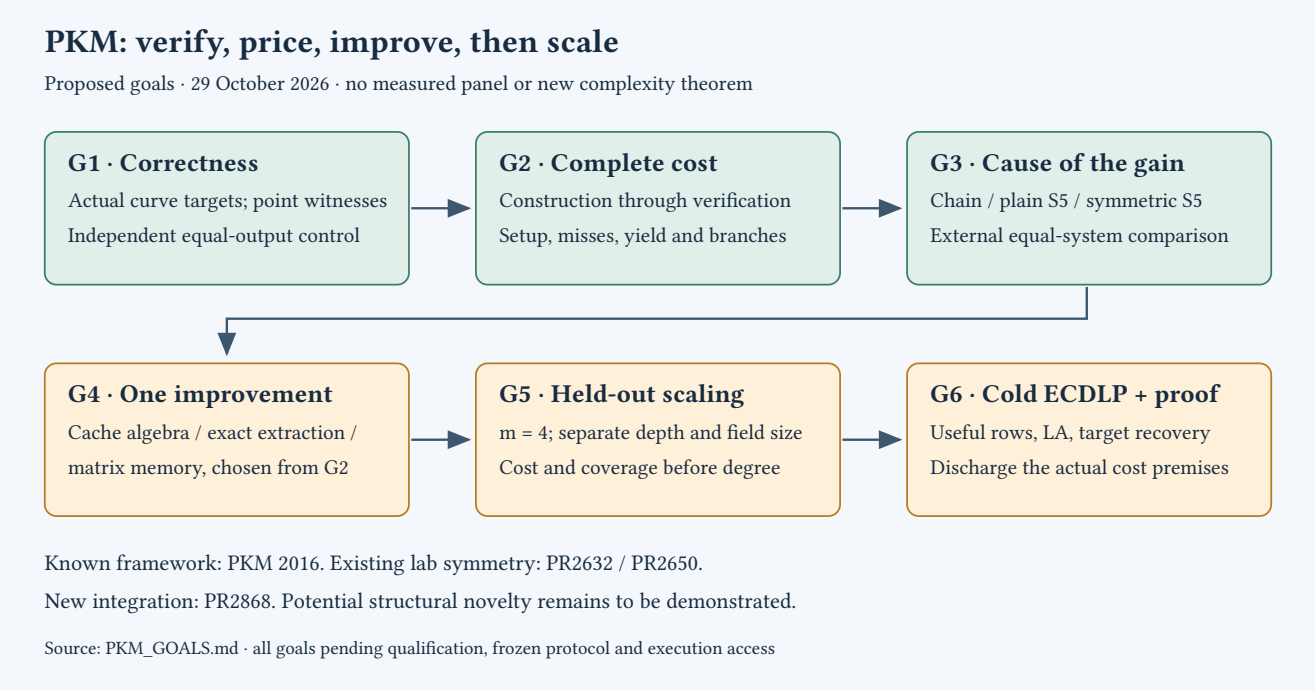
1. PKM: goals and novelty audit

29 October 2026 · proposed work program · no new measured result

The immediate question is whether symmetric PKM lowers the complete cost of a verified decomposition on actual curve targets. The later question is whether that moves a cold, single-target ECDLP. The detailed proposal and references are in `PKM_GOALS.md`; the blocked execution card is `backlog/XP-20261029-pkm-verified-oracle-program/`.



1.1. What is different

PKM's low-degree-map factor bases and summation-polynomial framework were published in 2016. Algorithm 1 explicitly accounts for permutations. This lab's elementary-symmetric tower and m=4 S5 instrument already appear in PR2632 and PR2650. PR2868 adds native point extraction and verification to that reviewed driver path, with explicit field-branch diagnostics and structural Lean proofs. It supplies an implementation bridge; no new ECDLP algorithm or complexity bound has been established.

The useful next contribution would be a complete, reproducible cost/yield comparison, followed by a verified improvement or a precise structural bound. Global novelty of the particular presentation has not been established.

1.2. What the older experiments already decided

The raw plain N=20 study reached productive degree-8 steps before a memory cap; the symmetric N=20 attempt aborted under 6 GB. Merely attempting N=20 is not a new experiment question. For symmetric m=4, degree 8 is already an input floor, so equal reported degree can conceal large cost growth. The old timer starts after constructing the equations. Construction, point extraction, failed queries and useful relation yield must be brought into the next comparison.

2. Measurable goals

Goal	Measure	Completion criterion
G1	Faithful point oracle	Zero lost/invalid supported decompositions against a native exhaustive tiny-curve control; equal domains and explicit unknowns.
G2	Complete cost and yield	Setup and every query phase priced or visibly unpriced; all successes verify; misses, caps, branches and valid yield retained.
G3	Presentation and solver ablation	Same E,V,Q across chain, plain S5 and symmetric S5; separate qualified external equal-system comparison.
G4	One measured improvement	Proposed: at least 20% less full cold oracle-batch cost on held-out data, with a paired interval excluding no improvement across the pricing band.
G5	Depth and field-size growth	m=4 first; predeclared completed points plus held-out check, or an explicit resource boundary. Separate fixed-p depth from field-size scaling.
G6	Cold ECDLP and the proof gap	Verified useful rows, factor-base logs and target recovery with every phase charged; faithful structural cost bound for any complexity claim.

2.1. First implementation candidates

1. Reuse validated curve/base algebra. Test a target-parameter representation of S5 against rebuilding the interpolation grid for every query. Charge cache generation, storage, cold setup and the declared reuse count.
2. Reuse triangular/quotient information for exact extraction. Make every full-field branch visible; an algebraic-only arm must report unsupported or capped extraction when it cannot proceed algebraically.
3. Reduce the largest measured matrix block and live basis. Change layout, ordering or tower-aware representation only with an equal-output ablation.

The memory objective is proposed as 2x lower peak memory with at most 10% extra complete cost, enabling another declared cell under the same cap. These are planning targets to freeze before data, not observed gains. Earlier F5/signature criteria already lost on degree and memory; repeating them needs a new reason. For the cost target, require point ratio at most 0.8, paired 95% upper ratio below 1, and unchanged or better completion and verified coverage.

2.2. Avoid the two main measurement traps

Planted positives and random field abscissae do not replace an unconditional public curve-target stream. Report natural valid yield and separate certified misses from resource-censored calls. At fixed p, increasing tower depth changes factor-base size and target consistency; that ladder cannot establish an ECDLP exponent. Matrix width, actual cost and output coverage accompany degree.

3. Decision boundary and execution

Under the existing rough model with quadratic-scale sparse LA, expected relation/rank yield and non-dominating setup, per-query cost r^γ gives balanced exponent $2\frac{1+\gamma}{m+1}$. At m=4, the arithmetic condition for that exponent to be below $\frac{1}{2}$ is exactly $\gamma < \frac{1}{4}$.

`lean/ThreeLeads/PKMGoalModel.lean` checks this arithmetic implication for rational model parameters only. It does not establish an oracle exponent, the LA/yield assumptions, curve coverage, a Rust cost bound or sub-rho ECDLP. The real proof goal is the precise symmetric ideal/filtration, its point-decomposition bridge and a cost bound on a quantified growing family. Small fixtures and degree plateaus are dependencies or observations, not that theorem.

3.1. Sequence and stop rules

G1–G3 first, then one justified G4 change, then G5. Attempt G6 once measured useful relations and accounting justify it. A correctness/verifier discrepancy stops the affected comparison. Report a null, loss or cap without changing the held-out set or creating a finite ratio from a censored baseline. Reject a candidate that misses its frozen target. Repeated caps trigger a specific memory/representation task. Finite negative scaling constrains that implementation/model extrapolation, not all PKM presentations.

The initial m=4 depth candidates are t=2,3,4, with t=5 conditional on validated sizing within an approved cap. Multiple primes and fresh curves/seeds need a frozen roster. The current driver is restricted to small primes below 2^{32}. Historical plain-arm 256–512 GB projections exceed the default pool's 128 GiB cap; this proposal requests no capacity increase.

3.2. Status

All scientific goals are pending. The native full-cost panel and replay verifier, qualified external arm, frozen protocol and supported execution/resume route are missing. Checkpoint/artifact access was unavailable. The backlog card is blocked and unfunded; no task, lease, objective or run is posted. The existing execution gates remain in force. Canonical performance and curve graphs are unchanged because the proposal adds no new measured value or curve.

3.3. Sources and limits

PKM primary text, sections 1,3–5: https://people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf . PR2632; PR2650 at `adde64fdc22d51dc180fe0b9f30a2d28adf992fe`; PR2868 at `8baeee2d446968936882adc503570fb957bb1429`; the PKM tower research note, sections 3.3–3.7 and rounds 1–8; `research/ic_next10_20261014/03-pkm-degree/README.md`; `research/prime_fall_degree_20261027/PREREGISTRATION.md`. APS ePrint 2017/609 returned HTTP403 in this audit. Full Yokoyama text remains unavailable. No exhaustive global-priority or universal negative claim is made.